Introduction to Spatial Data visualisation and Mapping in R

Statistics Netherlands (CBS)

Martijn Tennekes

Session 2: Methodology of Spatial Data Visualization

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Guidelines for creating maps

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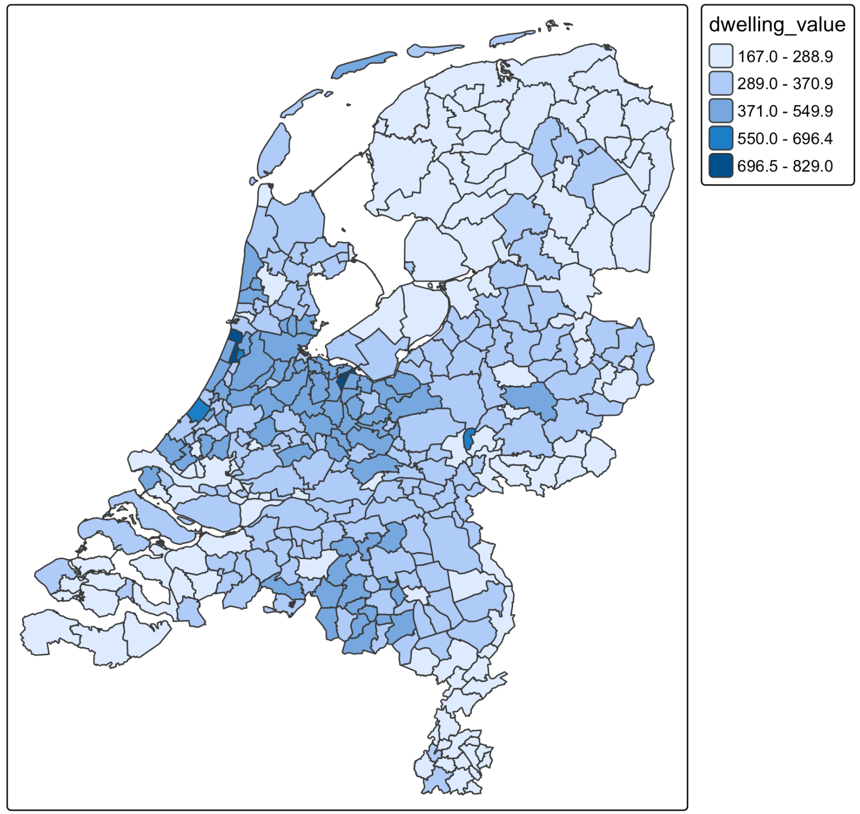
Content

- Standard thematic map types
  - → Choropleths
  - → Bubblemaps
- Advanced themtic map types
  - → Glyphs
  - → Cartograms
  - → Bivariate choropleths
- Exploring other map types
  - → Design space vs. taxonomy

Choropleth

```
tm_shape(NLD_muni) +
    tm_polygons(fill = "dwelling_value",
                fill.scale = tm_scale(style = "kmeans"))
```



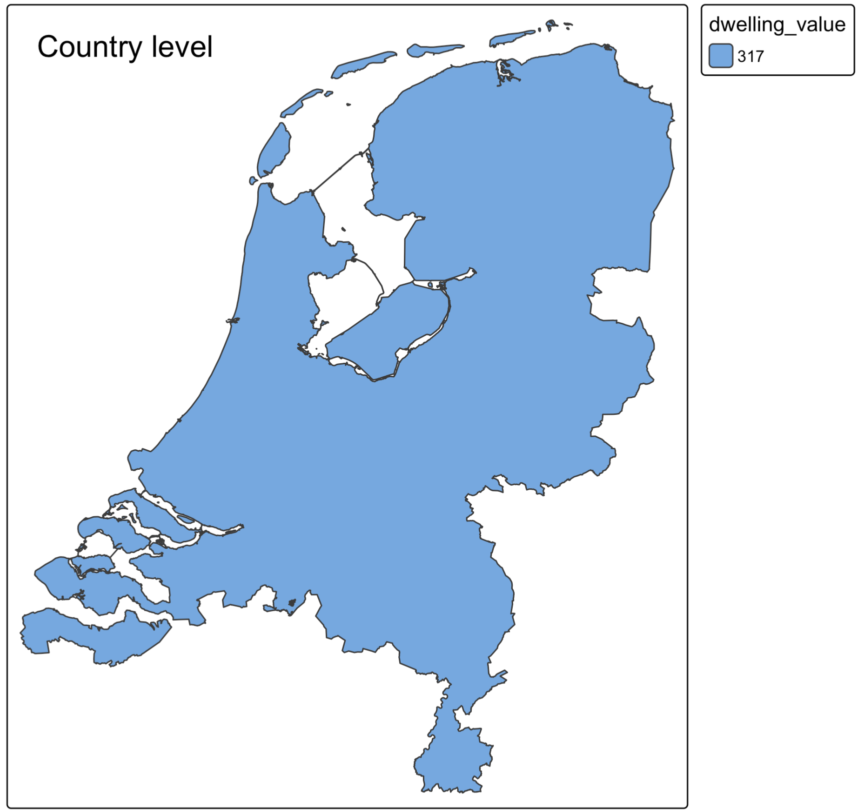
Check list

- What administrative level?
- Help people to read a map: borders, labels, zoomed-out map?
- What colors?
- Absolute or relative values?

What administrative level?

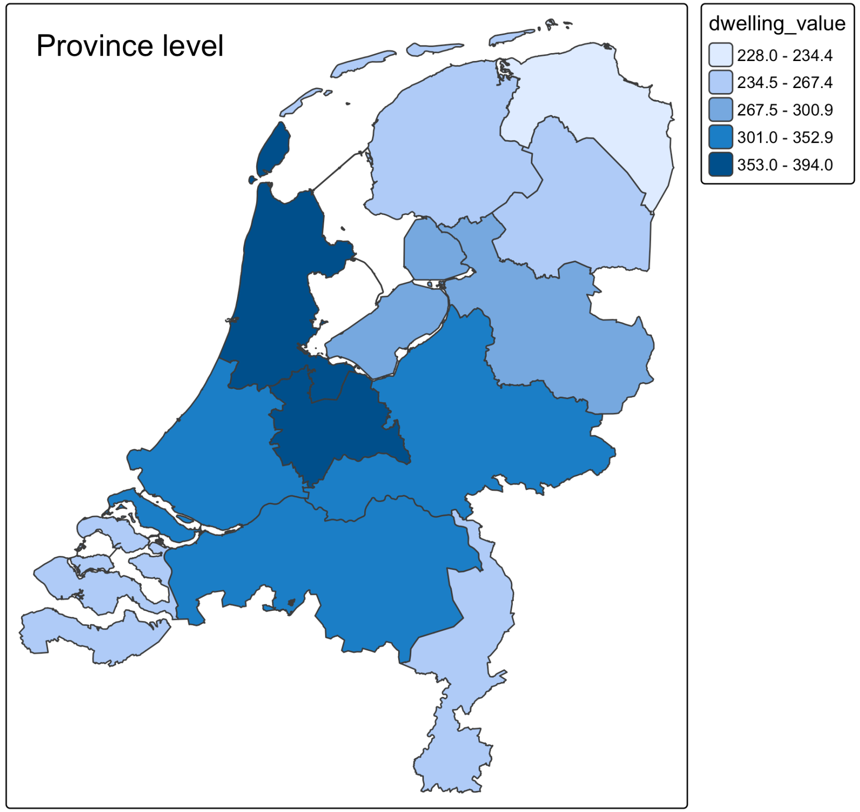
What administrative level?

```r
tm_shape(NLD) +
    tm_polygons(fill = "dwelling_value",
                fill.scale = tm_scale(style = "kmeans")) +
    tm_title_in("Country level")
```



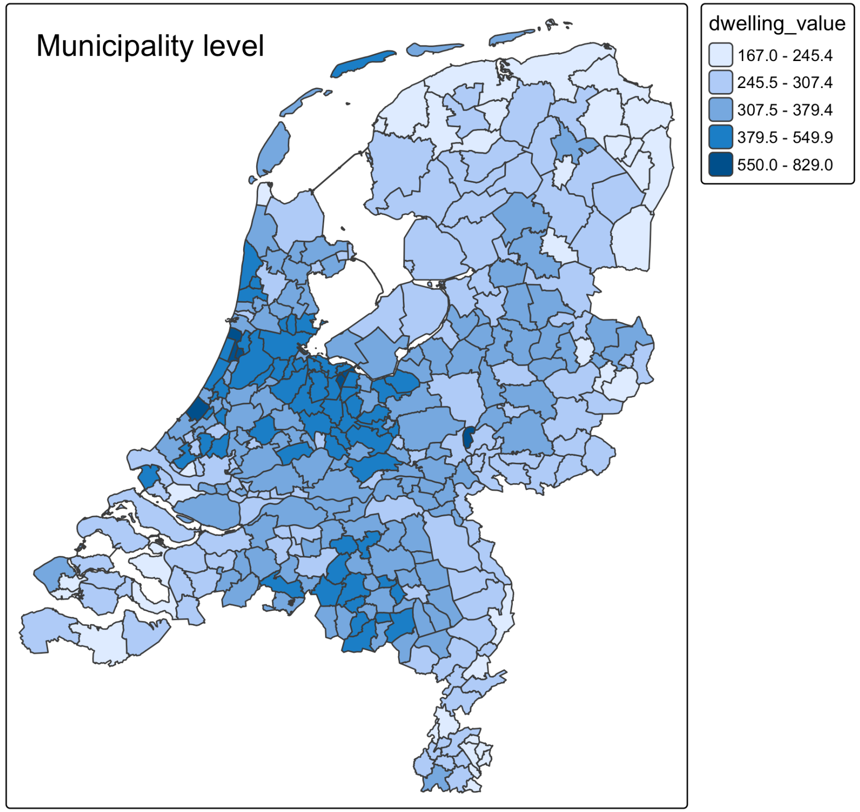
What administrative level?

```
tm_shape(NLD_prov) +
    tm_polygons(fill = "dwelling_value",
                fill.scale = tm_scale(style = "kmeans")) +
    tm_title_in("Province level")
```



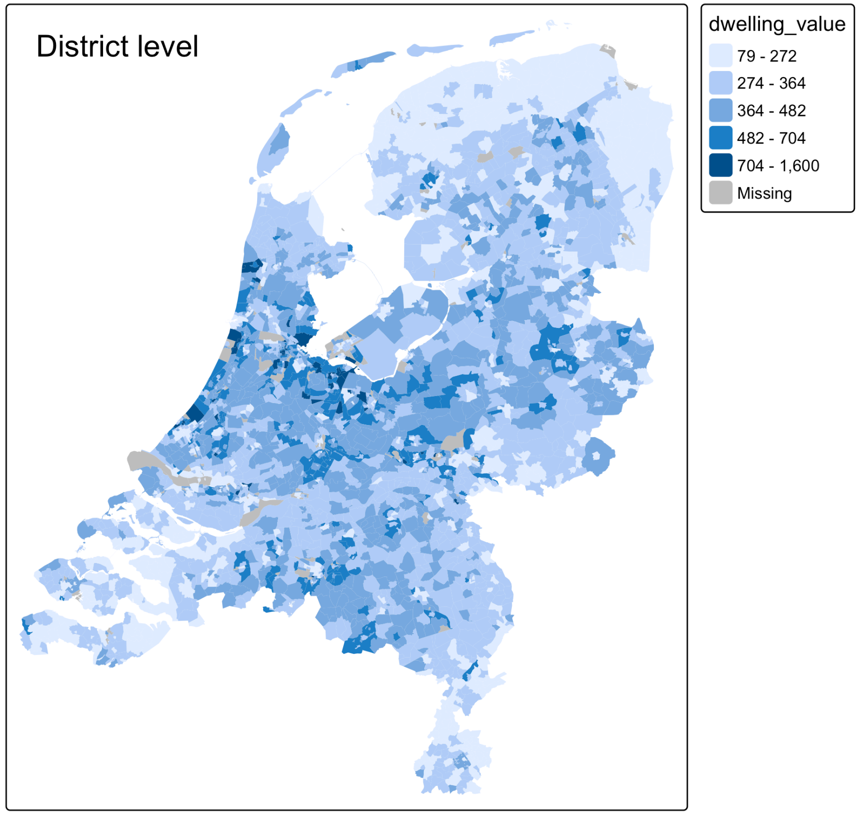
What administrative level?

```r
tm_shape(NLD_muni) +
    tm_polygons(fill = "dwelling_value",
                fill.scale = tm_scale(style = "kmeans")) +
    tm_title_in("Municipality level")
```



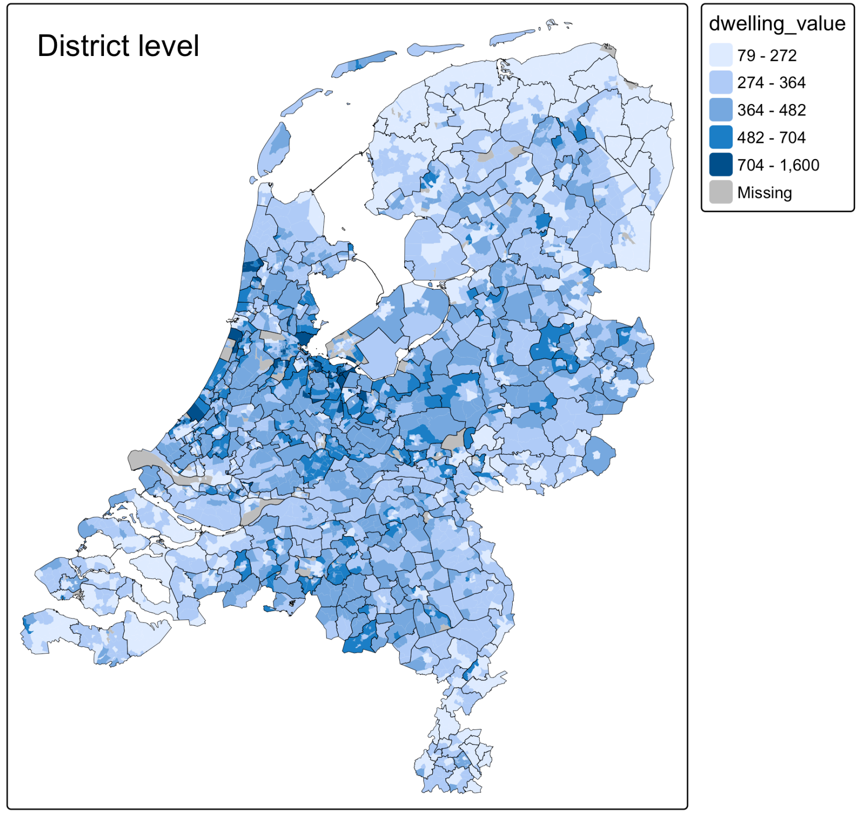
What administrative level?

```
(tm = tm_shape(NLD_dist) +
    tm_polygons(fill = "dwelling_value",
                col = NULL,
                fill.scale = tm_scale(style = "kmeans")) +
    tm_title_in("District level"))
```



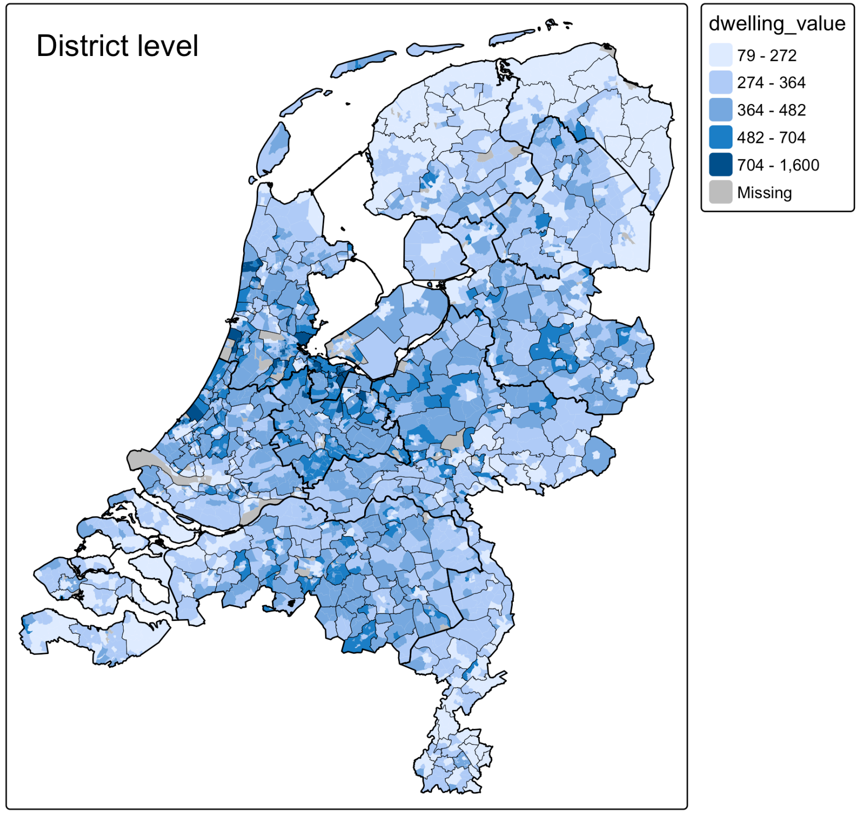
Geospatial information

```
(tm = tm + tm_shape(NLD_muni) +
    tm_borders(lwd = 0.3, col = "black"))
```



Geospatial information

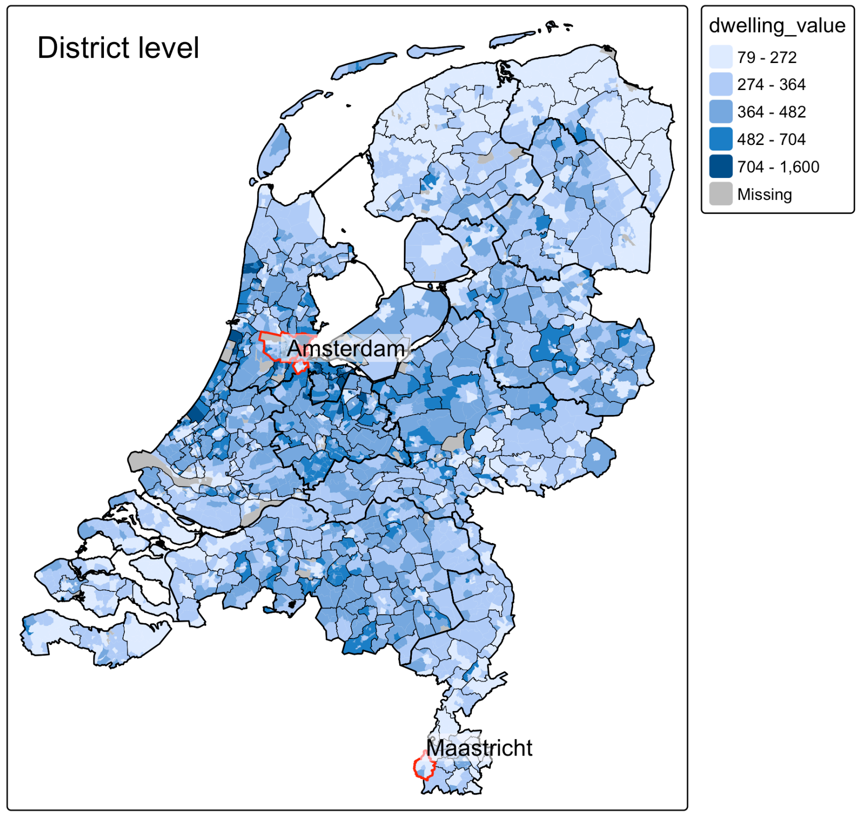
```
(tm = tm + tm_shape(NLD_prov) +
    tm_borders(lwd = 1, col = "black"))
```



13

Geospatial information

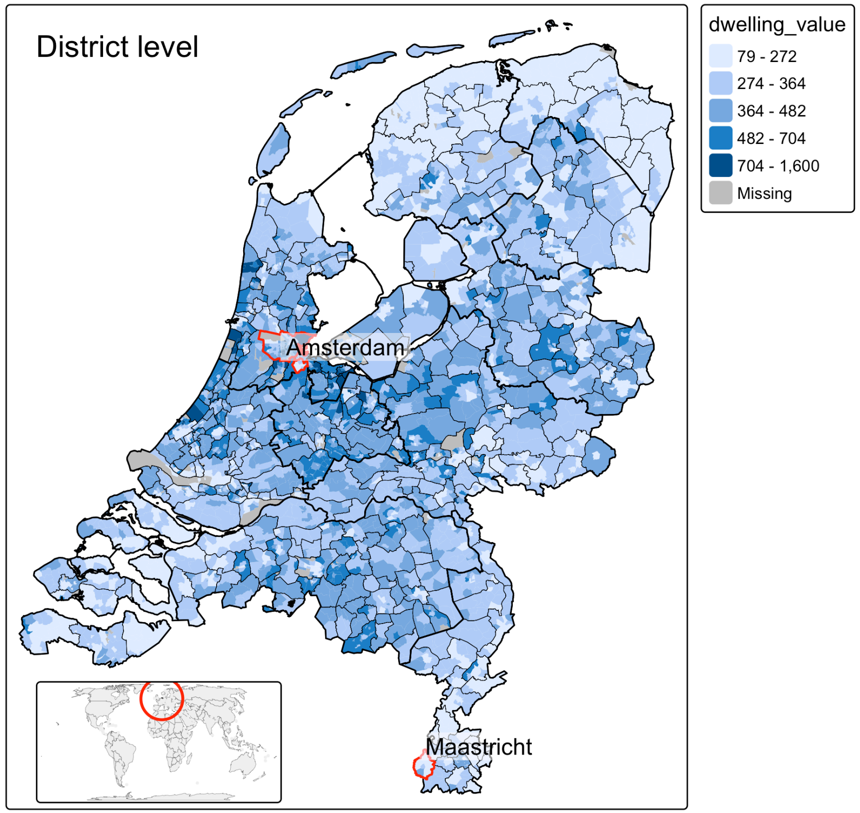
```r
set.seed(1) # for consistent label generation across maps
(tm = tm + tm_shape(NLD_muni) +
    tm_borders(lwd = 0.3, col = "black") +
    tm_shape(NLD_prov) +
    tm_borders(lwd = 1, col = "black") +
    tm_shape(NLD_muni |> filter(name %in%c("Amsterdam", "Maastricht"))) +
    tm_borders(col = "red", lwd = 1.5) +
    tm_labels("name", bgcol = "white", bgcol_alpha = 0.5))
```



Geospatial information

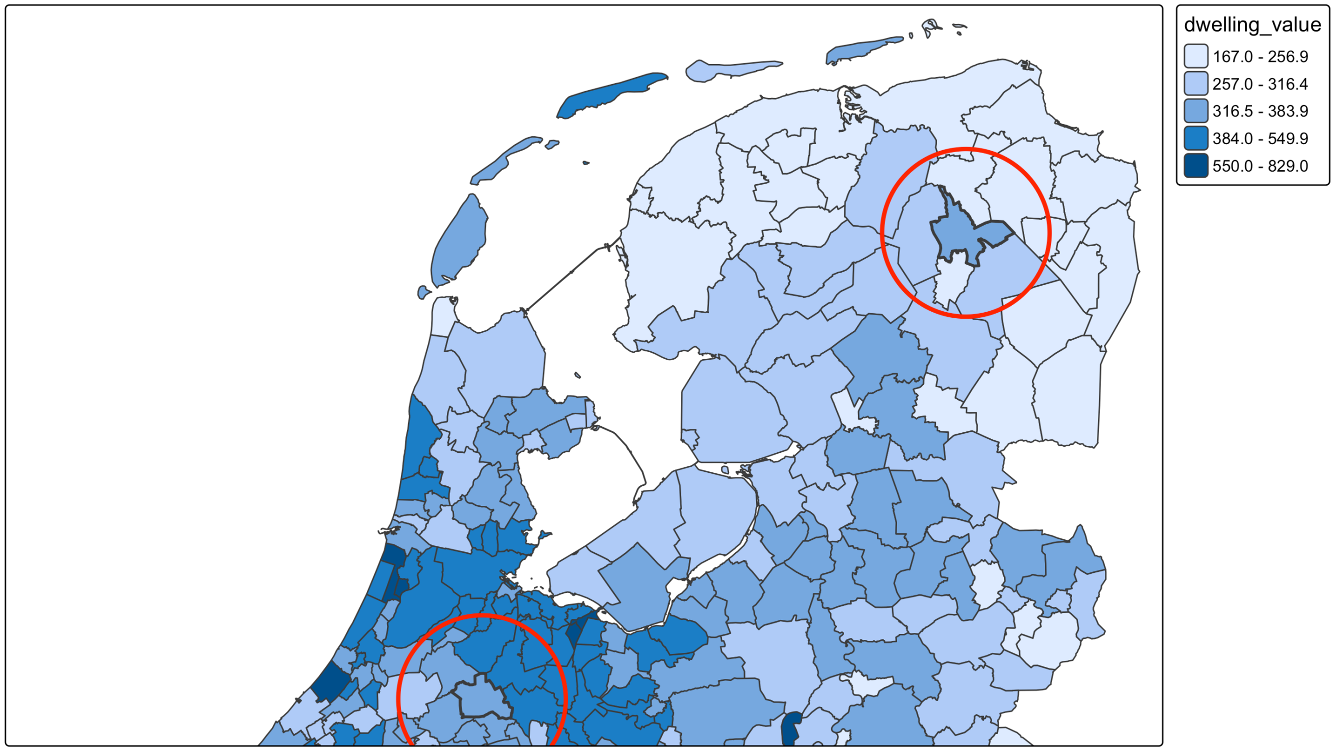
```r
set.seed(1) # for consistent label generation across maps
tm2 = tm_shape(World, filter = World$name == "Netherlands") +
    tm_polygons() +
    tm_bubbles(col = "red", size = 2, fill = NULL, lwd = 2) +
    tm_crs(crs = "auto")

tm
print(tm2, vp = grid::viewport(x = 0.3, y = 0.1, width = .3, height = .15))
```



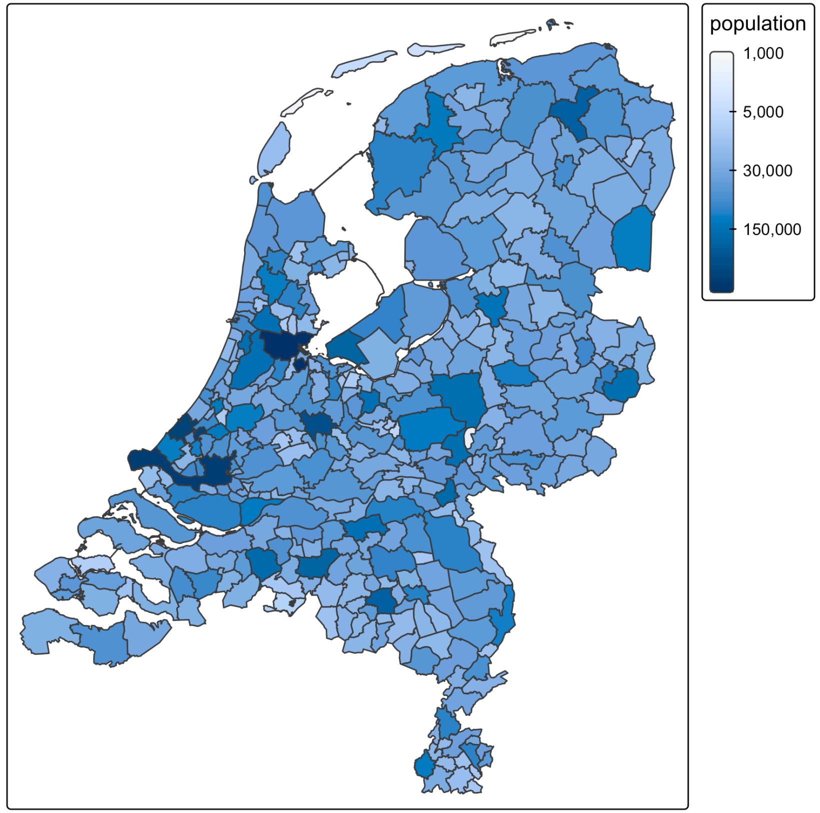
Choropleth: color is relative

```r
tm_shape(NLD_muni, bbox = sf::st_bbox(c(xmin = 13565, ymin = 450000, xmax = 278026, ymax = 619232)
    tm_polygons(fill = "dwelling_value",
                fill.scale = tm_scale(style = "kmeans")) +
    tm_shape(NLD_muni |> filter(code %in% c("GM1730", "GM0632"))) +
    tm_borders(lwd = 2) +
    tm_bubbles(col = "red", fill = NULL, size = 8, lwd = 3)
```



Choropleth: absolute numbers?

```
(tm = tm_shape(NLD_muni) +
    tm_polygons(fill = "population",
                fill.scale = tm_scale_continuous_pseudo_log()))
```



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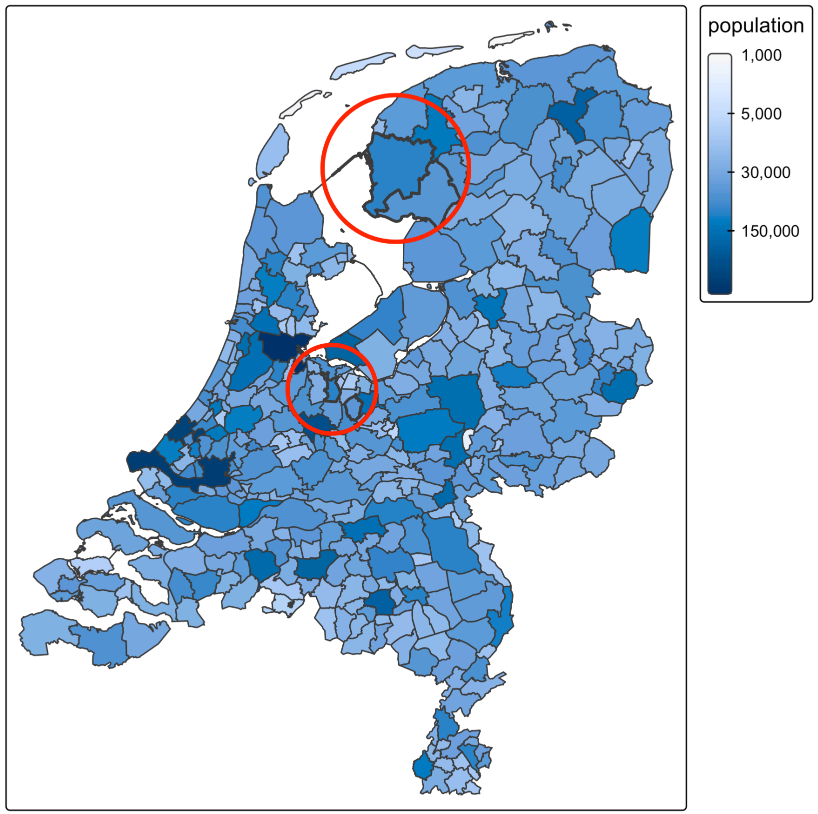
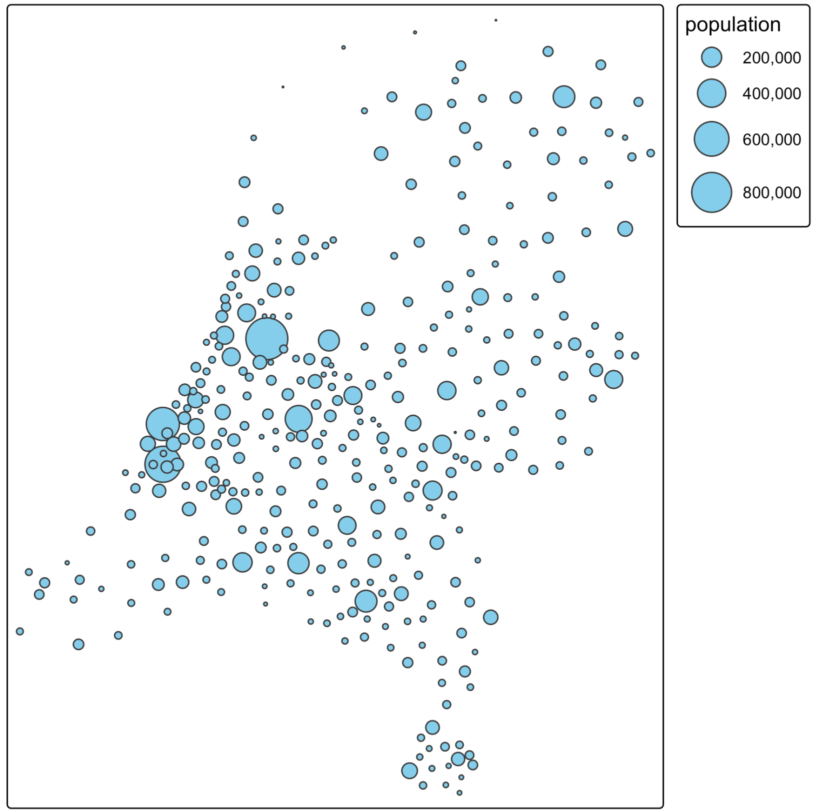
Choropleth: absolute numbers?

```
tm +tm_shape(NLD_muni |> filter(code %in% c("GM0402", "GM0342"))) +
    tm_borders(lwd = 2) +
tm_shape(NLD_muni |> filter(code %in% c("GM1900", "GM1940"))) +
    tm_borders(lwd = 2) +
tm_shape(NLD_muni |> filter(code %in% c("GM1900", "GM0402"))) +
    tm_bubbles(size = "area",
               size.scale = tm_scale_continuous(values.scale = 7, values = tm_seq(0, 1, power = 1/
               size.legend = tm_legend_hide(),
               col = "red",
               fill = NULL,
               lwd = 3)
```



Alternative: bubble map

```
tm_shape(NLD_muni) +
    tm_bubbles(size = "population",
               size.scale = tm_scale_continuous(values.scale = 2),
               fill = "skyblue")
```



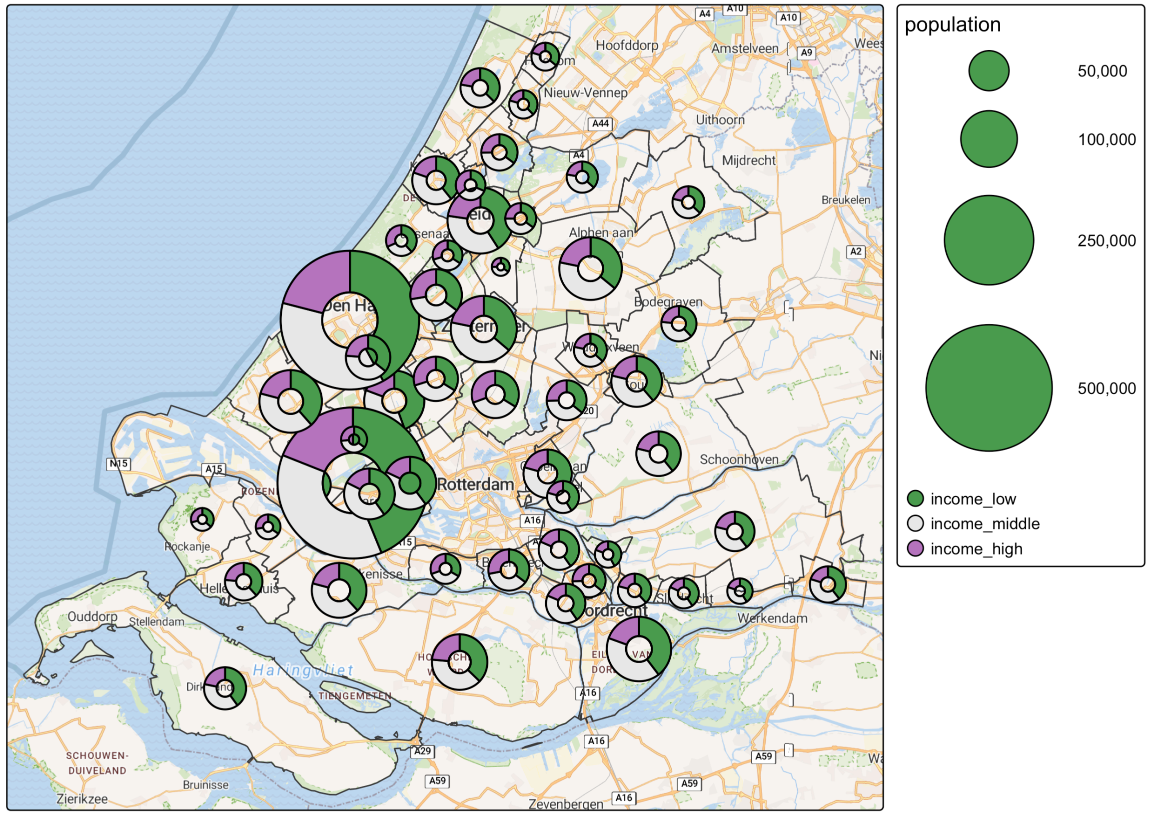
Proportional symbol map

- Spatial points are shown as symbols
- Symbols can be anything, but usually they are bubbles
- They are resized proportionally to a variable

Glyphs

- Instead of symbols, we can also draw small charts, called **glyphs**
- Enables us to plot **multivariate** point data

Glyphs

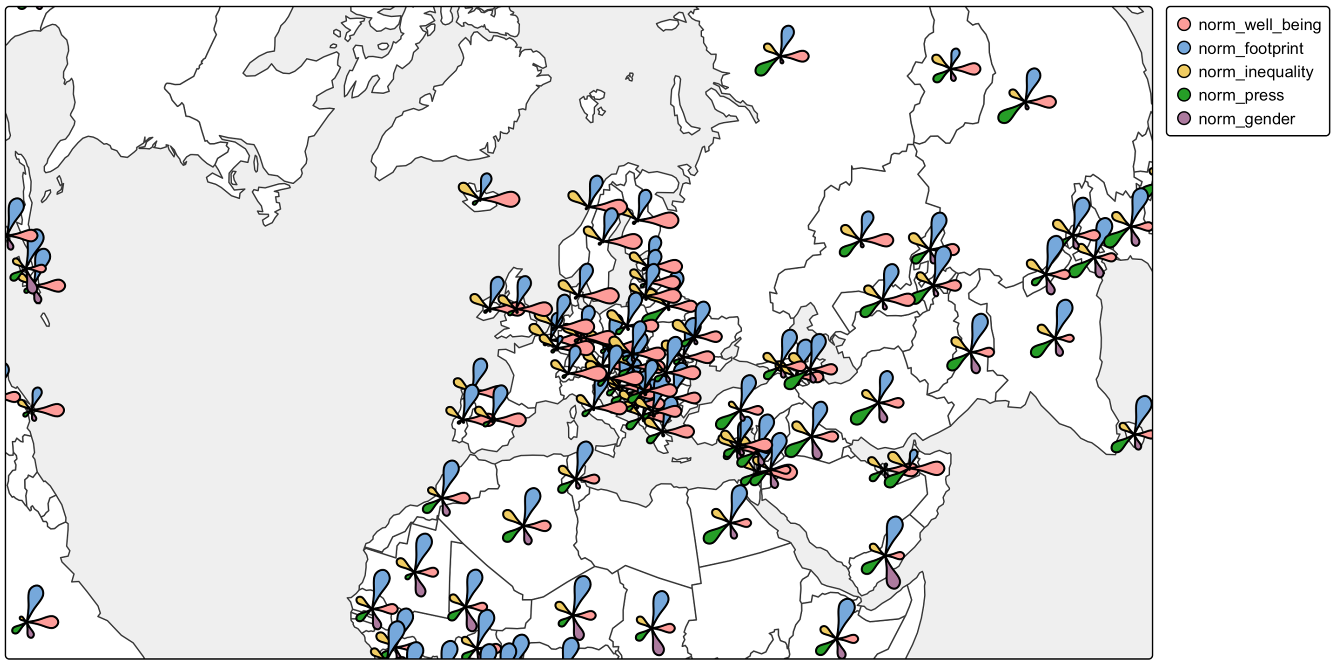

population
50,000
100,000
250,000
500,000
income_low
income_middle
income_high
Hoofddorp
Amstelveen
Nieuw-Vennep
Uithoorn
Mijdrecht
Breukelen
Bodegraven
Schoonhoven
Rotterdam
Rockanje
Oudddorp
Stellendam
Haringvliet
Werkendam
SCHOUWEN-
DUIVELAND
Zierikzee
Bruinisse
Zevenbergen

|---------|---------|---------|---------|
==

Flower glyphs

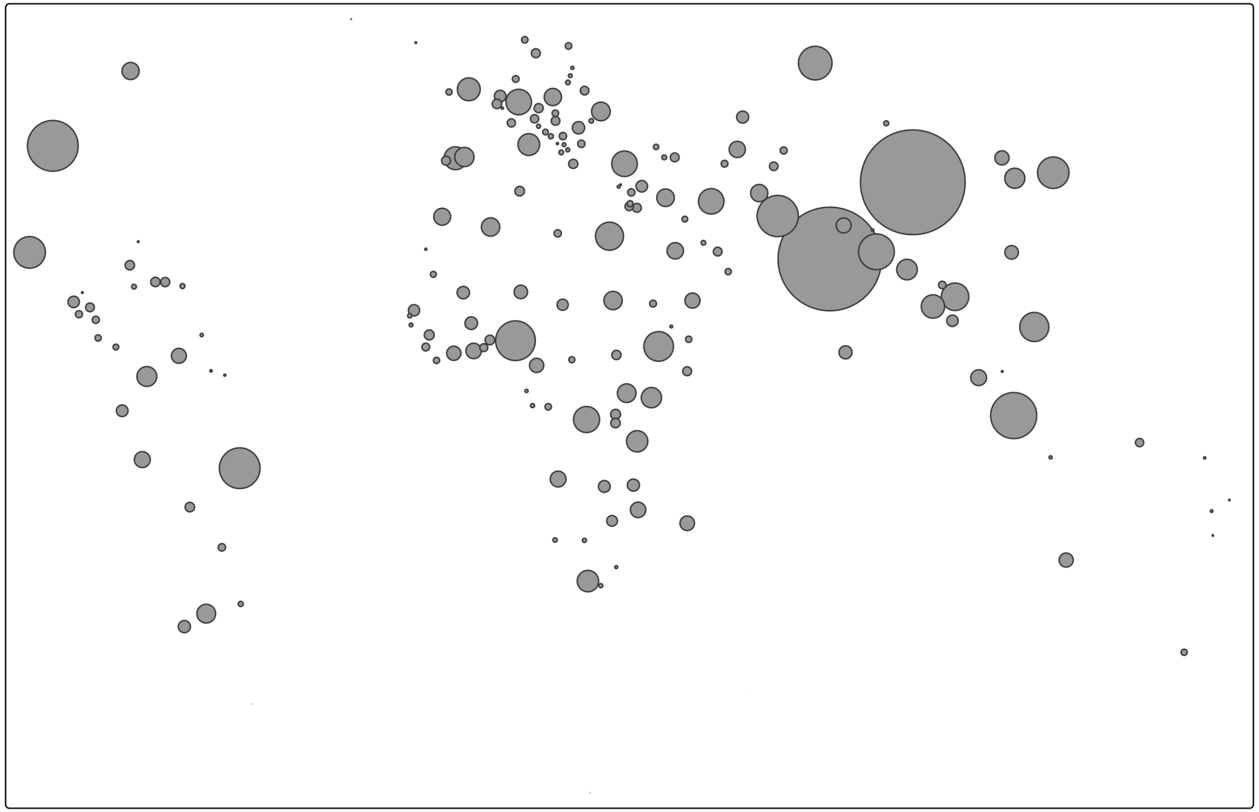
```
World$norm_well_being = World$well_being / 8
World$norm_footprint = (50 - World$footprint) / 50
World$norm_inequality = (65 - World$inequality) / 65
World$norm_press = (100 - World$press) / 100
World$norm_gender = World$gender

tm_shape(World, bbox = tmaptools::bb("Europe", width = 3, height = 1.5, relative = TRUE), crs = 30
    tm_polygons(fill = "white", popup.vars = FALSE) +
    tm_flowers(parts = tm_vars(c("norm_well_being", "norm_footprint", "norm_inequality", "norm_pre
               size = 0.6, popup.vars = c("norm_well_being", "norm_footprint", "norm_inequality",
    tm_layout(bg.color = "grey95")
```



Bubble map

```r
tm_shape(World, crs = "+proj=eck4") +
    tm_bubbles(size = "pop_est",
               size.scale = tm_scale(values.scale = 5),
               size.legend = tm_legend_hide())
```



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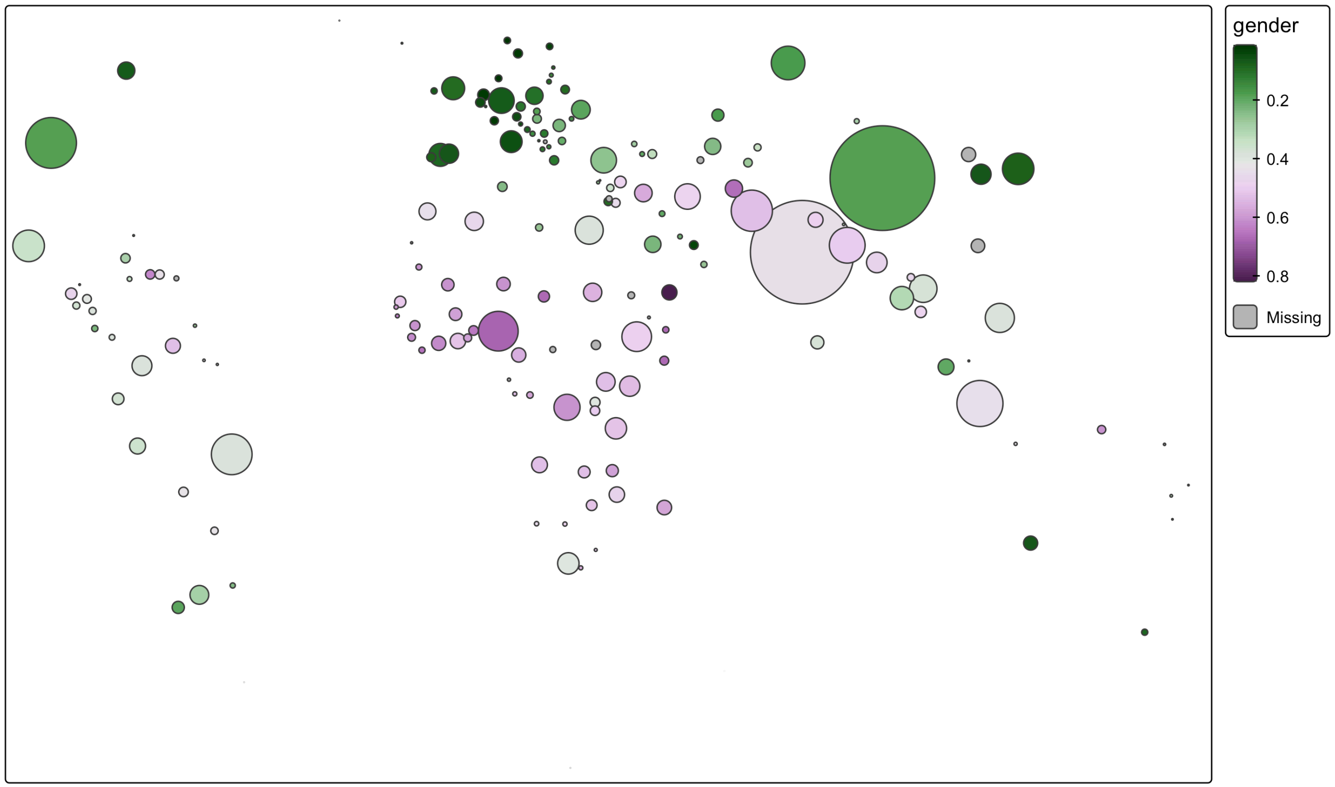
Bubble map (with data)

```
tm_shape(World, crs = "+proj=eck4") +
    tm_bubbles(size = "pop_est",
               size.scale = tm_scale(values.scale = 5),
               size.legend = tm_legend_hide(),
               fill = "gender",
               fill.scale = tm_scale_continuous(values = "-cols4all.pu_gn_div"))
```

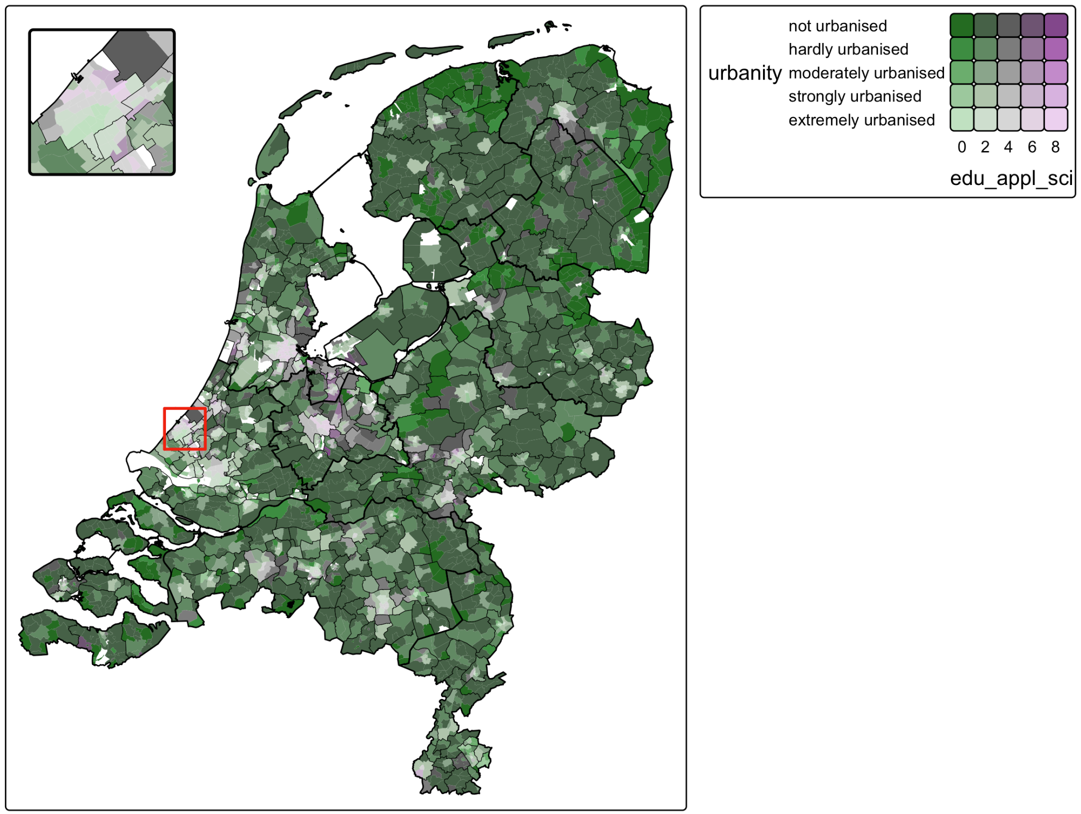


Dorling cartogram (with data)

```
tm_shape(World, crs = "+proj=eck4") +
    tm_cartogram_dorling(size = "pop_est",
                fill = "gender",
                fill.scale = tm_scale_continuous(values = "-cols4all.pu_gn_div"))
```

Bivarite choropleth

```
tm_shape(NLD_dist) +
    tm_fill(fill = tm_vars(c("urbanity", "edu_appl_sci"), multivariate = TRUE),
            fill.scale = tm_scale_bivariate(values = "-pu_gn_bivd"),
            fill.legend = tm_legend(col = "black")) +
tm_shape(NLD_muni) +
    tm_borders(lwd = 0.3, col = "black") +
    tm_shape(NLD_prov) +
    tm_borders(lwd = 1, col = "black") +
tm_inset(tmaptools::bb("The Hague"), position = tm_pos_in("left", "top"), height = 5, width = 5)
```



Taxonomy vs Design space

- Taxonomy for non-spatial data
- Design space for OD data

Donut maps

- Commuting in the Netherlands

Recap

- Choropleths and bubble maps are standard spatial visualization types
- For choropleths, beware of using absolute numbers
- Consider alternative methods, such as cartograms
- Choose color palettes with care